

Day 2 – gRNA design 2

In this course you will learn how to design gRNAs for two types of CRISPR/Cas9 editing experiments: whole-gene deletions (see gRNA design session 1) and precise mutations.

Part 2 – precision editing:

We'll design gRNAs and donor templates to engineer defined loss-of-function alleles by inserting stop codons into the open reading frame. But you can apply what you learn here to engineer any type of defined mutation, not just stop codons.

Engineering precise mutations requires additional design parameters beyond simple deletion gRNAs, including precise edit position relative to potential gRNAs and protection from re-cutting.

1. Go to your copy of the *galK* target region file on Benchling.
2. We want to introduce premature stop codons in the *galK* gene relying on repair by cellular HDR pathways. Whenever you want to make a specific mutation your first step is to annotate the position in the gene. Do this for the following mutations:

Groups 1 and 4	T7* (TAA)
Groups 1 and 4	G41* (TGA)
Group 2 and 5	T52*(TGA)
Group 2 and 5	G124* (TAA)
Groups 3 and 6	K101*(TAA)
Groups 3 and 6	A186*(TGA)

3. Use the Benchling gRNA design tool to design gRNAs for your two assigned mutations. It's easiest to pre-select the area your mutations lie in (I recommend selecting ca. 50 bp up and downstream of your mutation) and then start the design following the same steps as yesterday.
4. Select all your gRNAs and return to the 'SEQUENCE MAP' tab to see how they align relative to your target mutations. **Look at the different gRNAs and select one that you think would work well, motivate your choice.** Consider the design parameters we discussed.
5. Donor design: design two donor oligos following the schematic explained on the slides. To make your life easier you can first create an edited sequence and design the donor oligos there.